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The response of southern hemisphere saltmarsh plants and gastropods to experimental contamination by petroleum hydrocarbons

Peter J. Clarke* and Trevor J. Ward

CSIRO, Division of Fisheries, North Beach, W.A., Australia

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Abstract

Field experiments examined the response of two common perennial saltmarsh plants and their gastropod epifauna to the effects of weathered petroleum hydrocarbons. Application of the petroleum hydrocarbons to the saltmarsh mimicked an accidental spill, but confined the contamination to small areas. Populations of the perennial chenopod, *Sarcocornia quinqueflora*, and the perennial grass, *Sporobolus virginicus*, showed little inertia and senesced rapidly after the application of weathered Bass Strait crude oil and diesel (1 $l \cdot m^{-2}$). No resprouting from underground stems or recruitment of seedling was detected up to 17 months after the application of hydrocarbons to *Sarcocornia*. Dead stems of the perennial grass *Sporobolus* persisted in areas treated with oil and diesel for up to 12 months, but showed no signs of resprouting from basal shoots or rhizomes originating from culms within the treated areas. Slow recovery from rhizomes originating outside the plots was evident after a few months but tiller growth appeared to be inhibited by residual hydrocarbons. No recruitment of seedlings was observed in the denuded plots and no other macrophytes were observed to colonise these areas until the end of the study (17 months). The response of saltmarsh gastropods to petroleum hydrocarbons shows greater inertia and stability than the vascular plants. Initial mortality was high, but migrations from the edges of the treated areas restored densities to control and pre-treatments levels within a few months. The reduction in cover of plants apparently had little effect on the abundance of gastropods although residual effects of the hydrocarbons may have inhibited predators of gastropods from the openings created by the death of saltmarsh plants. We predict that the widespread contamination of saltmarshes in south-eastern Australia by spills of crude oil or diesel would result in the loss of vegetation cover and reductions in the abundance of gastropod epifauna.

Key words: Crude oil; Field experiment; Gastropod; Hydrocarbon spill; Saltmarsh

* Corresponding author. Botany Department, University of New England, NSW 2351, Australia.

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1. Introduction

Spills of petroleum hydrocarbons are often stranded in the intertidal zone of estuaries and consequently saltmarshes and mangroves are prone to frequent contamination (Gundlach & Hayes, 1978). These spills (mostly crude oil) on marine ecosystems attracts considerable public attention and larger oil spills have provided the impetus for a variety of scientific investigations, mostly related to northern hemisphere biota. Studies of the impacts of petroleum hydrocarbons on saltmarsh communities fall into three broad approaches: Experimental studies in glasshouses (e.g. Ferrell et al., 1984; Scholten et al., 1987; Li et al., 1990), comparative studies in the field (e.g. Cowell, 1969; Stebbings, 1970) and experimental studies in the field (e.g. DeLaune et al., 1979; Baker, 1979; McGuinness, 1990). Research on the effects of petroleum hydrocarbons on intertidal macrophytes have produced a wide range of results and conclusions. Grant (1991) attributes this to temporal and spatial heterogeneity of natural systems and the skills and advocacy of researchers. The conclusions of Grant's critical review of the experimental methodology used to study effects of petroleum hydrocarbons on mangroves are also applicable to many saltmarsh studies. These were; (1) lack of replication, (2) treatments not approximating real spills, and (3) statistical analyses poorly described. Some of these problems also reflect the more widespread inadequacies of some ecological research highlighted in the reviews of Hurlbert (1984) and Underwood (1981).

There have been few reported accidental spillage's of petroleum hydrocarbons into saltmarshes in the southern hemisphere (e.g. Anink et al., 1985), although there have been many instances where crude oil has washed into nearby mangroves (McGuinness, 1990). The only previous study of the impact of hydrocarbons on southern hemisphere saltmarshes is that of McGuinness (1990) who examined effects of Dubai light crude oil on mangrove and saltmarsh macro-invertebrates in the field. McGuinness (1990) used weathered oil and applied the oil in a way which mimicked a real spill. However, the removal of residual oil from the sediment a few hours after its application may have been inappropriate because, in a typical spill, crude oil can persist as a free layer for several days.

Australian saltmarshes fall into three broad groups: those on tropical coasts, those experiencing a Mediterranean climate and those on temperate coasts (Adam, 1990). The latter are characterised by a dominance of the perennial chenopod herb *Sarcocornia quinqueflora* (Ung.-Sternb.) A. J. Scott adjacent to mangroves and taller rushes and sedgelands higher on the shore (Adam, 1990). The macro-invertebrate fauna are mainly detritivores or consumers of microflora and are dominated by gastropods and crabs (Kaly, 1988).

Our study examined the effects of experimental spills of petroleum hydrocarbons on two common saltmarsh vascular plants and the associated gastropod epifauna in temperate southeastern Australian saltmarshes. A hydrocarbon spill was simulated in a manner similar to that of Grant et al. (1993), by using enclosures which allowed the pre-weathered hydrocarbons to rise and fall with the tide until the hydrocarbon slick no longer persists on the water surface. We tested the null hypothesis that there is no effect of petroleum hydrocarbons on populations of the biota tested.

2. Materials and methods

2.1. Study sites

Study sites were located in two tidal creeks Currambene Creek (patch 1) and Moona Moona Creek (patch 2), entering Jervis Bay on the south coast of New South Wales, Australia (Fig. 1). Sites were selected in undisturbed saltmarsh vegetation adjacent to mangroves at about 1.2 m above tidal datum (spring tidal range 1.8 m). The mangroves consist of stands of *Avicennia marina* var *australasica* (Walp.) Moldenke and *Aegiceras corniculatum* (L.) Blasco. About 36 vascular plant species are found in the saltmarsh (Clarke, 1993). The saltmarsh adjacent to the mangrove zone usually consists of monospecific patches of the grass *Sporobolus virginicus* (L.) Kunthand the chenopod subshrub *Sarcocornia quinqueflora*. Other common saltmarsh species found in the lower saltmarsh of Jervis Bay include *Samolus repens* Pers., *Wilsonia backhousei* J.D. Hook. *Triglochin striata* Ruiz Lopez & Pavon and *Sclerostegia arbuscula* (R. Br) P.G. Wilson.

Seventeen species of gastropods, three species of bivalves, and seven species of crustaceans have been recorded in saltmarshes of Jervis Bay (Hutchings, pers. comm.). The most common gastropods were *Tatea* spp., *Assimenia bucciniodes* Tenison-Woods, *Salinator solida* (von Martens), *Ophicardelus* spp., *Bembicium auratum* (Quoy & Gaimard) and *Littorina luteola* (Quoy & Gaimard). Three species of crab, *Heloecius cordiformis* (H. Milne Edwards), *Helograpsus haswellianus* (Whitelegge), and *Sesarma erythrodactyla* (Hess) also commonly occur in the saltmarsh.

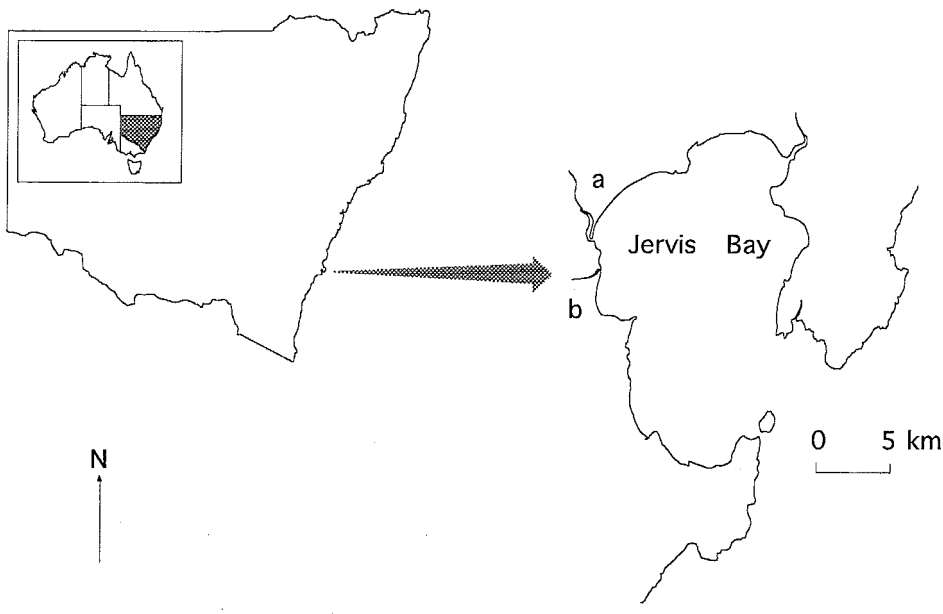


Fig. 1. The location of the study sites (a) Currambene Creek, (b) Moona Moona Creek in Jervis Bay, NSW, Australia.

2.2. Application of petroleum hydrocarbons

Hydrocarbons were contained in clear perspex enclosures until a slick was no longer present on the incoming water surface (four consecutive high tides over a spring tide period). The enclosures were open to tidal rise and fall via a U-bend buried in the sediment but connecting the interior and the exterior of the enclosure so that escape of surface-bourn hydrocarbons was minimal. Weathered hydrocarbons were applied at the peak of a spring high tide so that on the ebb tide they coated the saltmarsh and its fauna. The experiment comprised four treatments: weathered Bass Strait crude oil (a light crude oil but with a high wax content), weathered diesel fuel, dispersant (Corexit NSC 6850), and crude oil plus dispersant. Two controls were also used: a perspex enclosure control and an open, unboxed control. The crude oil and diesel were weathered for 24 h on seawater in plastic bins before application at 1 $l \cdot m^{-2}$, thus most low weight hydrocarbons would have evaporated (McAuliffe, 1976). The dispersant was applied to the weathered oil in the enclosure at a ratio of 1:20 when treatments were applied in the field. Each treatment was randomised and replicated four times in each of the two widely separated saltmarshes (patches), boxes were spaced ≈ 2 m apart.

2.3. Saltmarsh flora

Cover of two saltmarsh plants (*Sarcocornia quinqueflora* and *Sporobolus virginicus*) was estimated from photographs of experimental sites taken before (4.12.90), 2 months (8.1.91) and 8 months (17.7.91) after treatment. Photographic slides were projected onto a 625 point grid frame and 50 random points were sampled. A pilot study determined the minimum number of points to resolve cover differences of about 5%. In most cases, this was 50 random points. Estimates of cover were arc-sine transformed before statistical analysis. A two-factor ANOVA was used to analyse these data; treatments were fixed while patches were random. Additional counts of tillers were also made for *Sporobolus virginicus* up to 17 months after treatment.

2.4. Saltmarsh fauna

Prior to application of treatments the common gastropod epifauna of the experimental areas were counted in quadrats measuring 0.25 m^2. Areas containing *Sarcocornia* were not sampled because searching for gastropods would damage the brittle stems of the plants. Small gastropods (*Tatea* spp. and *Assiminia*) were not counted because *Sporobolus* would be excessively disturbed in searching for these animals. Four gastropods were counted, *Salinator solida, Ophicardelus* spp., *Bembicium auratum* and *Littorina luteola*. It was not possible to distinguish between the three species of *Ophicardelus* in the field so they were lumped and analysed as a group. Counts of live individuals of the four gastropod species were made in each treatment area and in controls 24 h after the application of treatments and thereafter at approximately 1-month intervals for 12 months. A pilot study before oiling indicated that a coefficient of variation of 0.10 could be achieved with a quadrat size of 0.25 m^2 and four replicates. After the animals were counted, they were replaced in situ. Disposable gloves were initially worn to pick animals from experimental areas with hydrocarbons so that

Table 1
ANOVA of cover of live *Sarcocornia quinqueflora*

Factor	df	Before			After 1 month			After 8 months		
		MS	*F*	*p*	MS	*F*	*p*	MS	*F*	*p*
Treatment	5	0.0016	2.53	n.s.	0.0042	4.30	–	0.0041	10.95	<0.01
Patch	1	0.0034	3.29	n.s.	0.0019	7.21	<0.05	0.0018	8.59	<0.01
T×P	5	0.0006	0.61	n.s.	0.0009	3.66	<0.01	0.0003	1.79	n.s.
Error	36	0.0010			0.0002			0.0002		

Data transformed to arc-sine; $n=4$ replicates in each of two patches; n.s. not significant, $p>0.05$.

treatments were not cross-contaminated. A two-factor ANOVA was used to analyse total gastropod densities and densities of each of the four groups; treatments were fixed while patches were random. Time was not incorporated into the analysis because samples were repeatedly taken from the same sample unit leading to the possibility of temporal non-independence.

3. Results

3.1. Saltmarsh flora

Live cover of the saltmarsh herb *Sarcocornia quinqueflora* were very similar among experimental areas before treatments began (Table 1). Treatments of oil (OI), oil and dispersant (OID), and diesel (DL) reduced live cover in comparison with controls, and there was no recovery 8 months after treatment (Fig. 2). Further observations made 12

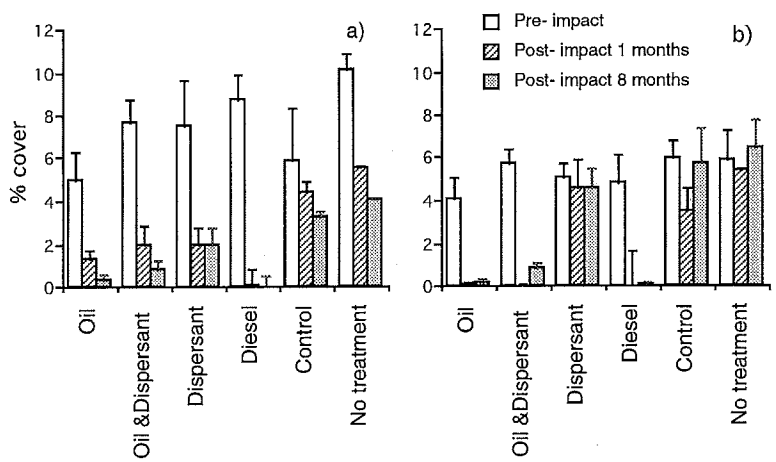


Fig. 2. Mean (SE) live cover for *Sarcocornia quinqueflora* before, 1 and 8 months after application of hydrocarbons in two patches (a & b) in Currambene Creek.

Table 2
ANOVA of cover of live *Sporobolus virginicus*

Factor	df	Before			After 1 month			After 8 months		
		MS	*F*	*p*	MS	*F*	*p*	MS	*F*	*p*
Treatment	5	0.0544	3.02	n.s	0.7776	16.31	–	0.5502	43.31	–
Patch	1	0.1722	7.46	<0.01	0.0040	0.35	n.s.	0.0232	5.22	<0.05
T×P	5	0.0180	0.78	n.s.	0.0475	4.16	<0.01	0.0127	2.85	<0.05
Error	36	0.0230			0.0114			0.0044		

Data transformed to arc-sine; $n=4$ replicates in each of two patches; n.s. not significant, $p>0.05$.

and 17 months after application of treatments showed no seedling recruitment of *Sarcocornia quinqueflora* in any of the treatments nor was there any resprouting of underground stems in sites treated with oil, oil and dispersant, or diesel.

Significant differences in the live cover of the saltmarsh grass, *Sporobolus virginicus,* were detected among experimental patches prior to the application of treatments (Table 2), but these differences were small (Fig. 3). Application of hydrocarbons led to a much larger change in live cover (Fig. 3). Live cover was reduced when treated with OI, OID and DL in comparison with controls and there was little or no recovery 8 months after treatment (Fig. 3). A significant interaction term indicated that patches responded differently to the treatments, but this only seems to apply to the dispersant treatment (cf. Fig. 3a with 3b). Whereas cover estimates show the virtual elimination of live material oiled areas the tiller counts indicate that some tillers survive (Fig. 4). Diesel treatment, however, eliminated tillers for up to one year, after which a few tillers emerged from rhizomes originating outside the treated area.

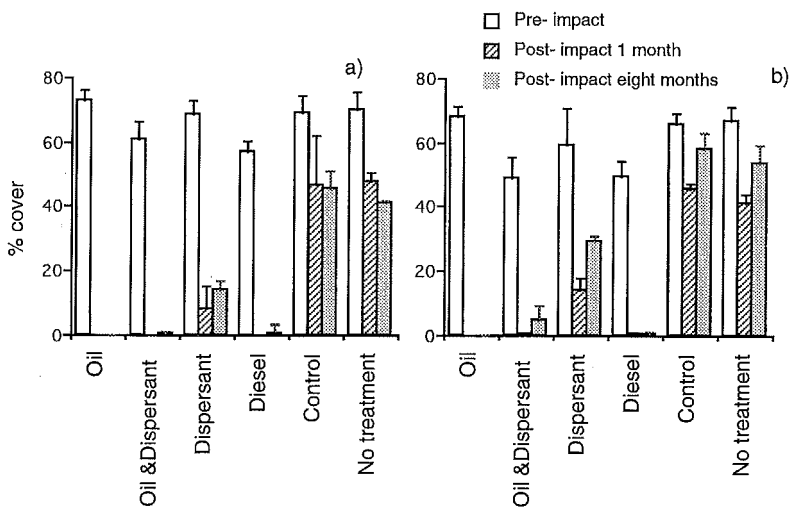


Fig. 3. Mean (SE) live cover for *Sporobolus virginicus* before, 1 and 8 months after application of hydrocarbons. (a) Currambene Creek, (b) Moona Moona Creek.

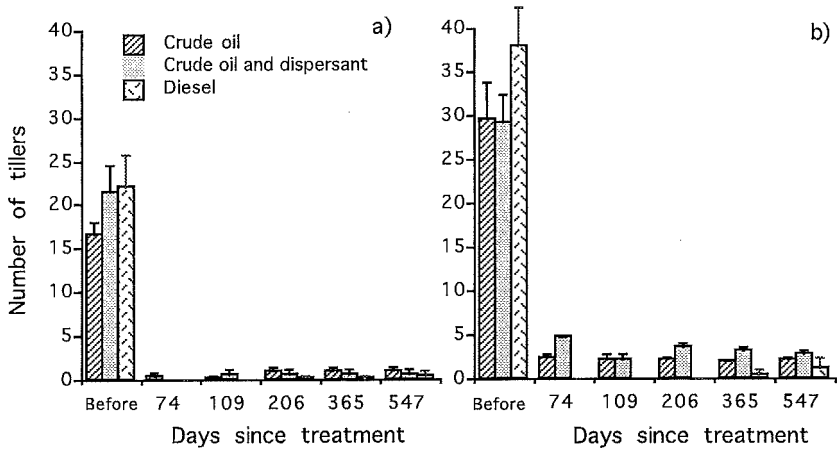


Fig. 4. Mean (SE) number of *Sporobolus virginicus* tillers present after application of petroleum hydrocarbons. (a) Currambene Creek, (b) Moona Moona Creek.

Table 3
ANOVA summary table for density of *Bembicium* (*Bem*) and *Littorina* (*Lit*)

Source	df	Before		10 days		38 days		74 days		177 days		365 days	
		Bem	*Lit*	*Bem*	*Lit*	*Bem*	*Lit*	*Bem*	*Lit*	*Bem*	*Lit*	*Bem*	*Lit*
Treatment	5	n.s.	n.s.	<0.01	<0.01	n.s.	<0.01	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.
Cochran's test		n.s.	n.s.	n.s.	<0.05	n.s.	<0.05	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.

$n = 4$ replicates in one patch, n.s. not significant, $p > 0.05$.

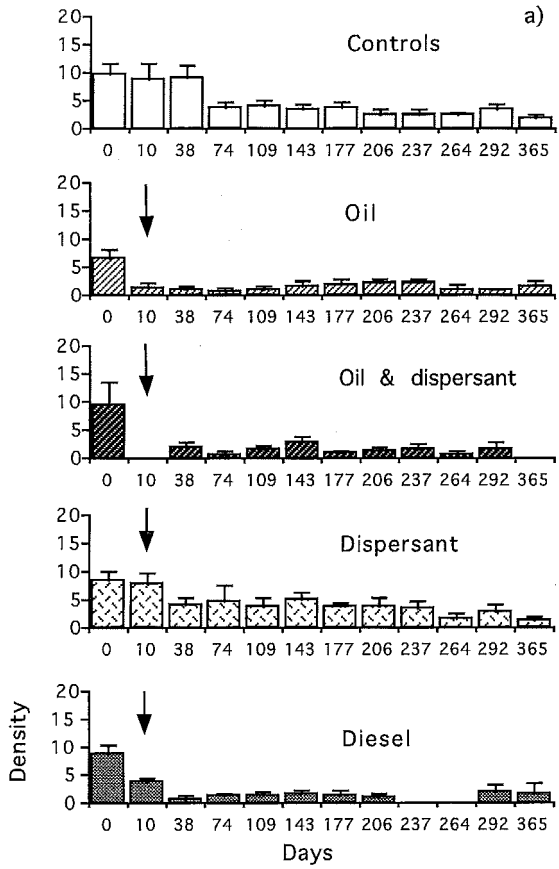
3.2. Saltmarsh fauna

Preliminary analysis for heterogeneous variances (Cochran's C) showed that variances were generally unstable for most taxa immediately after treatments were applied, thus it is difficult to assess the significance of treatment and patch effects as they may represent type I errors (Tables 3 and 4). Prior to the application of treatments there

Table 4
ANOVA summary table for density of *Salinator* (*Sal*) and *Ophicardelus* (*Oph*)

Source	df	Before		10 days		38 days		74 days		177 days		365 days	
		Sal	*Oph*	*Sal*	*Oph*	*Sal*	*Oph*	*Sal*	*Oph*	*Sal*	*Oph*	*Sal*	*Oph*
Treatment	5	n.s.	n.s.	–	n.s.	<0.05	<0.05	<0.05	<0.05	n.s.	n.s.	n.s.	n.s.
Patch	1	<0.001	<0.001	<0.001	<0.005	n.s.	n.s.	<0.05	n.s.	<0.01	<0.05	n.s.	<0.01
T×P	5	n.s.	n.s.	<0.05	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	<0.01
Cochran's test	n.s.	n.s.	<0.05	<0.05	<0.05	<0.05	n.s.	n.s.	n.s.	n.s.	n.s.	n.s.	

$n = 4$ replicates in each patch,n.s. not significant, $p > 0.05$.



were no significant differences in abundance of gastropods among areas to be treated within patches, although the abundances of *Ophicardelus* and *Salinator* differed between patches (Tables 3 and 4). After the application of the various treatments significant main effects or interactions were detected for all gastropods (Tables 3 and 4). Widespread mortality of gastropods and some crabs was noticeable in all areas treated with O, OID, and DL immediately after treatment (next day).

Bembicium and *Littorina* were not sufficiently abundant in both patches to be analysed in a two factor model. Instead a single factor model was employed using data from patches where they were sufficiently abundant. In these patches both species decreased in abundance in response to OI, OID and DL 10 and 38 days after their application, but thereafter no significant differences could be detected (Table 3 and Fig. 5). *Salinator* and *Ophicardelus* taxa were sufficiently abundant in both patches for them to be analysed with patches as a factor. Both taxa responded initially in the same way as *Bembicium* and *Littorina* with a significant decrease in numbers for OI, OID and DL treatments, thereafter abundances appear to recover to the levels of the controls (Table 4 and Figs. 6 and 7). Twelve months after the application of treatments the

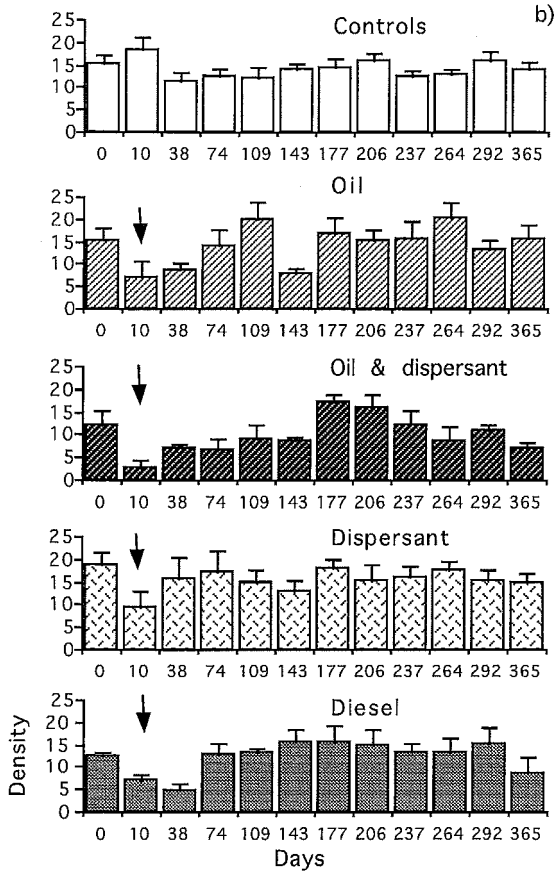


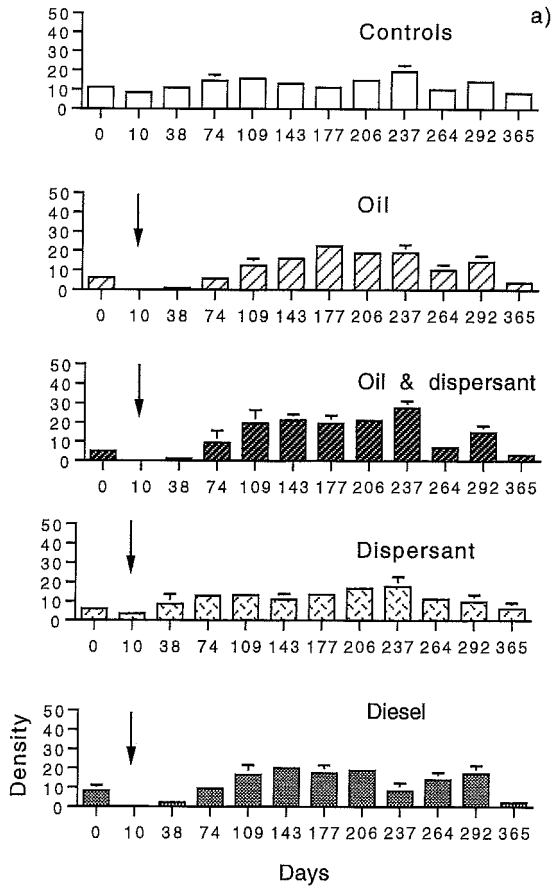
Fig. 5. Mean (SE) density of (a) *Littorina* and (b) *Ophicardelus* in experimental treatments and pooled controls. Arrow indicates application of treatments.

number of gastropods in one patch appears to decreased relative to the controls (Figs. 6 and 7).

4. Discussion

4.1. Saltmarsh plants

The methods used in our study differ from most field experiments on saltmarsh plants because we used weathered petroleum products and applied them in a way that closely resembles a real spill. In this regard our results are not directly comparable other studies. Similarly, the amounts of crude oil used (1 $l \cdot m^{-2}$) are higher than those used in the widely cited studies of Baker (0.45 $l \cdot m^{-2}$). Our results contrast with previous studies because both perennial species were killed by petroleum hydrocarbons (Bass Strait crude and diesel), and recovery, at least for 17 months, appears to be slow. Both



species appear to have little inertia (*sensu* Underwood, 1989) with senescence occurring within a few weeks. The perennial chenopod herb, *Sarcocornia,* showed no sign of recovery, either by vegetative encroachment from the edge of the areas or by seedling recruitment, for 17 months after application of weathered crude oil and diesel fuel. Also, no other vascular plants colonised these areas in this time.

Dead stems of the perennial grass *Sporobolus* persisted in areas treated with oil and diesel for up to 12 months but show no signs of resprouting from basal shoots or rhizomes originating from culms within the treated areas. Slow recovery from *Sporobolus* rhizomes originating outside the plots was evident after a few months although tillers only achieved low densities in the treated plots and appeared to be inhibited by residual hydrocarbon effects. No seedling recruitment was observed in the treated plots and no other vascular plants were observed to colonise these areas until the end of the current observations (17 months).

Studies of northern hemisphere saltmarsh plant species show a wide response to differing types and amounts of petroleum hydrocarbons (Baker, 1979; Hershner &

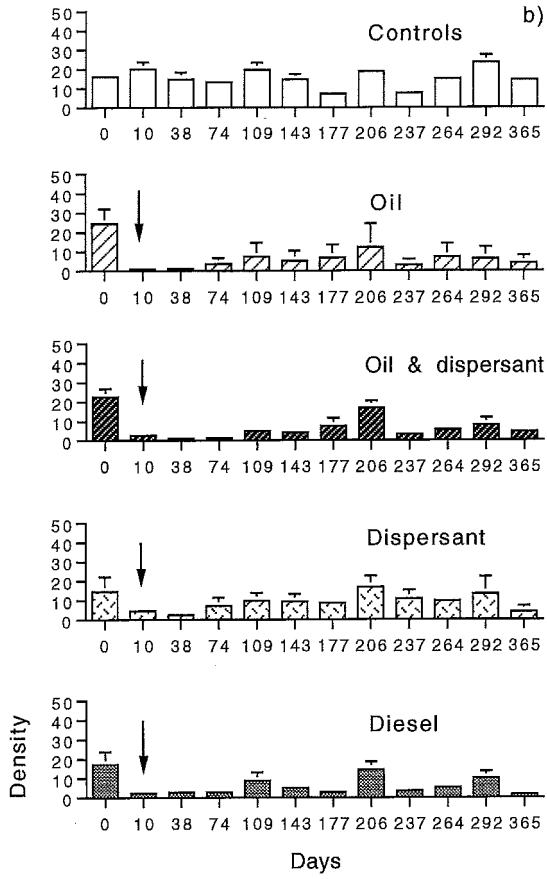
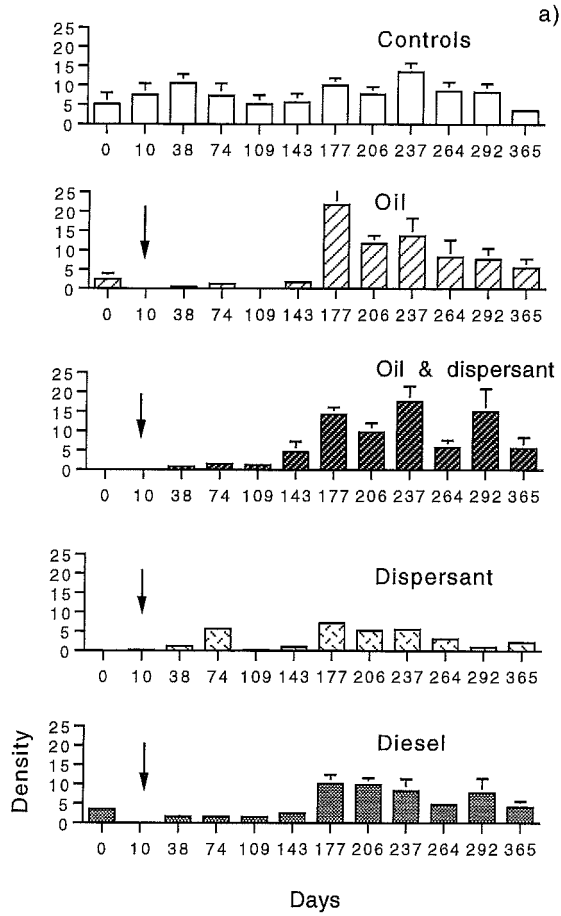


Fig. 6. Mean (SE) density of *Salinator* in (a) Currambene Creek, (b) Moona Moona Creek for experimental treatments and pooled controls. Arrow indicates application of treatments.

Lake, 1980; Ferrell et al., 1984; Scholten & Leendertse, 1991). Of those species studied in field experiments the annual chenopods appear to be the most susceptible, while fast growing perennial grasses recover from repeated oilings and in some instances growth is enhanced (Baker, 1979; Li et al., 1990). Our results suggest that the slow growing perennial grass (*Sporobolus virginicus*) and the perennial chenopod (*Sarcocornia quinqueflora*) rank among the most sensitive saltmarsh plants to spills of hydrocarbons. Both species are common in the low saltmarsh of south-eastern Australian and often form monospecific stands. These results are similar to observations of the response of temperate Australian saltmarshes to physical perturbations from vehicles and trampling (Adam, 1991) and indicate that these saltmarshes may be very susceptible to disturbance from petroleum hydrocarbons and that recovery from such disturbances could be slower than the rates reported for northern hemisphere saltmarshes. In part, this stems from the absence of species with a fast opportunistic growth and the scarcity of annual species in low saltmarshes of southeastern Australia.



4.2. Macroinvertebrates

The macroinvertebrates showed greater inertia and stability in their response to petroleum hydrocarbons than did the vascular plants. Gastropods showed high initial mortality, but migrations from the edge of the treated areas restored densities to control and pretreatments levels within a few months. The reduction in density of *Ophicardelus* relative to the controls a year after the initial impact may reflect a residual effect of the hydrocarbons, although this effect was not consistent among the two patches. The reduction in cover of plants apparently had little initial effect on the abundance of gastropods which is surprising in view of the results of Kaly (1988) who found that the removal of *Sarcocornia* cover decreased the abundance of common gastropods. Several studies have suggested that removal of plant cover might increase the activity of predators and decrease the abundance of the saltmarsh fauna (Kaly, 1988; Warren, 1985; Vince et al., 1976). Reduction in plant cover may not have affected the initial

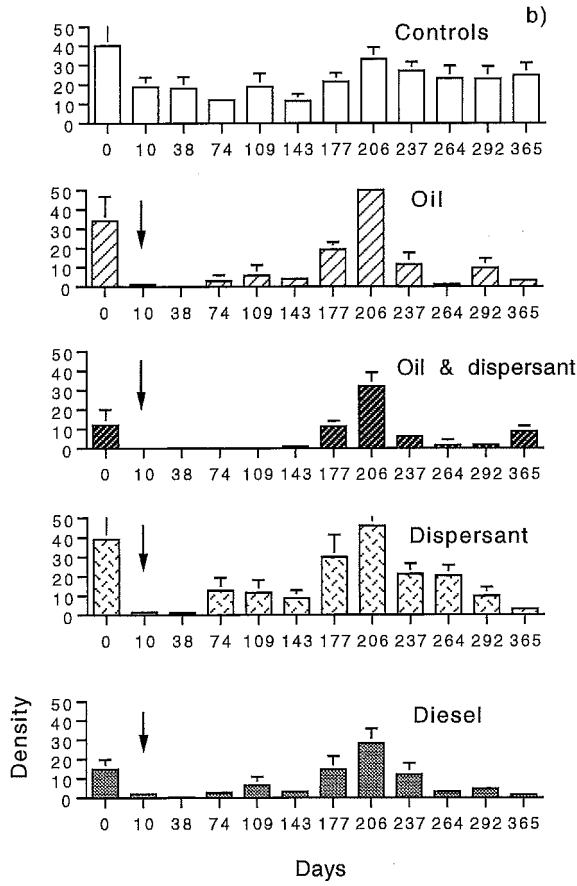


Fig. 7. Mean (SE) density of *Ophicardelus* in (a) Currambene Creek, (b) Moona Moona Creek for experimental treatments and pooled controls. Arrow indicates application of treatments.

abundance of gastropods in our experiment because the residual hydrocarbons may have also discouraged predators for up to a year. However, this effect may have worn off after 12 months and increased predation may account for reductions in *Salinator solida* in treated areas.

McGuinness (1990) found no evidence of reduced abundances of gastropods at several sites in mangroves in Botany Bay an area previously contaminated with crude oil. His experiments in mangroves and in *Sarcocornia*saltmarsh showed that application of weathered Dubai oil (1 $l \cdot m^{-2}$) caused short term reductions of some gastropods but there was little indication of long-term residual effects even after a repeated oiling. His experimental application of crude oil differs from ours in that 30–60% of the residual-free oil was removed from the sediment after application, nevertheless, the response of the gastropods in our study is similar. It appears that the initial dose of oil has most impact and that residual oil has few direct effects on the gastropods

studied. McGuinness (1990) qualified the results of his field experiments in three ways; (1) effects of a "mousse" spill could be different to that of a thin film, (2) residence time of the oil might be longer and (3) size of patches impacted was small. The first and last qualification also apply to our results. In particular, the small size of contaminated patches in our experiment may have underestimated the response of gastropods and other macroinvertebrates because of immigration from nearby unaffected saltmarsh. In a large spill immigration from unaffected areas would be much slower and recovery may rely on plankton settlement rather than immigration as discussed by McGuinness (1990). Juveniles of these gastropods recruit into the same areas as adults sporadically throughout the year (Kaly, 1988), and the lack of saltmarsh cover after a spill may affect larval settlement and subsequent recruitment.

Acknowledgements

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